

Theories of perception

Main path
Detailed concepts

Such as

Such as

2 opposed accounts



Bottom-up

= Stimulus-guided
= Data-driven

Top-down

= Knowledge-guided
= Concept-driven

Basic assumptions

Basic assumptions

- **Perception = sensation** (determined by retinal image)
- Perception occurs **automatically**
- **No cognition** involved
- Perception is **directly measurable**
- Perception is effortless: **fast and accurate**

Sensation
Uninterpreted sensory impressions created by detection of a stimulus

Perception
Processes to make sense of sensations

- **Perception > sensation** (retinal image needs to be interpreted)
- Perception is **not automatic**
- **Cognitive processing** involved
- Perception is **not effortless**
- Perception can be **erroneous** or **equivocal**, e.g., **illusions/ambiguous figures**

Main critique

Main critique

- Perception is **not always accurate** (retinal image is not sufficient for perception)
e.g., **illusions, ambiguous figures**

Visual illusions

- Constructed (e.g., Fraser spiral)
- In the real world (e.g., moon illusion)

Ambiguous figures

- Figure/ground ambiguity (e.g., vase/faces)
- Feature ambiguity (e.g., duck/rabbit)
- Depth ambiguity (e.g., Necker cube)

Real world ambiguity

- Inverse projection problem (too many 3D objects produce same 2D retinal image)
- Superimposed objects
- Unusual views

- Effortful perception **does not match subjective experience**
- **Illusions persist** despite knowledge

Theories

Theories

- **Structuralism**
- **Psychophysics**
- **Direct perception (Gibson)**

- **Gestalt Psychology**
- **Constructed perception (Gregory)**

In detail

Structuralism

= Describing elements of consciousness

- Perception = **Sum** of sensory elements
- Wilhelm **Wundt**, 1879
- Edward **Titchener**, 1901
- Method = **Introspection**
- **Critique:** Subjective method

Definition

Conscious experience can be broken down into basic elements, which can then be combined to describe all human experiences

Definition

Method to reduce perception into its basic elements

Psychophysics

= Measuring elements of perception

- Perception = **Measurable** sensation
- Method = **Threshold measurements**

>> **Measurement of the absolute threshold**
Gustav **Fechner**, 1860

Definition

Just noticeable intensity threshold to detect a stimulus

Method of adjustment
(participant adjusts stimulus intensity)

Returns

Intensity needed to perceive a stimulus at a moment in time
e.g., dark adaptation experiments

Method of limits
(trials of ascending/descending stimulus intensity)

Returns

Mean intensity required for a stimulus to be perceived
i.e., stimulus is perceived or not

Method of constant stimuli
(many intermixed trials of different stimulus intensities)

Returns

Probability with which stimulus is perceived at a particular intensity
e.g., 50% chance level at which stimulus is perceived

>> **Measurement of the difference threshold**
Ernst **Weber**, 1834

Definition

Just noticeable intensity difference to tell two stimuli apart

Weber's Law
= The difference threshold is a constant proportion of the respective stimulus

This means

The stronger stimulus 1, the greater the change required for stimulus 2 to be perceived as different

>> **Magnitude estimation (scaling)**
Stanley **Stevens**, 1957

Definition

Relation between (objective) stimulus intensity and (subjectively) perceived intensity

- **Response compression**
(perceived intensity < stimulus intensity)
- **Response expansion**
(perceived intensity > stimulus intensity)

- **Critique:** Valid methods, but unclear how meaning is extracted from sensation

Direct Perception (James Gibson, 1966)

= **The environment delivers everything** necessary for perception

- Perception should be investigated in natural environment = **ecological approach**
- Direct perception = **direct sensation**
- **Basic statement**
Perception takes place in the **optic array** and is directly based on the **invariant information** in the visual field which is extracted by the **observer's movement**
- Perception through **object affordance** = objects tell us what they are and what they are used for
- **Critique:** Cannot explain erroneous perception

Definitions

Optic array = light in the environment

Observer's movement
Makes optic array ambient (light changes) revealing invariant information

Invariant information, e.g.,

- Optic flow pattern
- Texture gradient
- Horizon ratio
- Gravity
- Straight lines
- Reflectance of areas

Sensory information from the retina is insufficient, **perception is a matter of interpretation**

This means

Constructive Perception (Richard Gregory, 1970)

= active and interpreted perception

- **Indirect perception:** Interaction between sensation & cognition (e.g., knowledge, expectation, context, motivation)
- Perception > Sensation
- **Basic statement**
Perception is hypothesis-guided on the basis of stored knowledge
- Based on **individual factors** (thus error-prone)
- Perception through **experience**
- **Critique:** Explains interpretation of sensation, how we perceive and recognise objects (i.e., on the basis of stored knowledge about objects), but cannot explain why illusions are not learnt

This means

Known rules about the world are applied to extract meaning from sensation

In terms of

Grouping & Segmentation (Segregation)

According to the following Gestalt **principles**

- Proximity
- Similarity
- Common fate
- Good continuation
- Closure
- Relative size
- Surroundedness
- Horizontal/vertical orientation
- Asymmetry

→ **Law of Prägnanz** determines which principle is applied
"Of several geometrically possible organisations that one will actually occur which possesses the best, simplest, and most stable shape"

Gestalt Psychology

= **Perceptual organisation** to achieve Gestalt (whole configuration/object)

- **Basic statement**
The whole is more than the sum of its parts
i.e., Perception > Sensation
- **Critique:** Great heuristics (problem solving strategies), but purely descriptive, no real theory